Luvna Dhawka

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Education

2023–present **The University of North Carolina at Chapel Hill**
PhD in Bioinformatics and Computational Biology
Advisor: Dr. Natalie Stanley

2016–2020 **Colgate University**
B.A. in Molecular Biology, *cum laude*
Minor in Mathematical Systems Biology

Research Interests

Understanding how neuroimmune cells communicate across the brain and periphery in neurodegenerative diseases. Combining cell-cell communication analysis with machine learning to uncover disease-associated immune signatures and potential biomarkers.

Research Experience

2023–Present **Graduate Student Research Assistant** | UNC Chapel Hill
Stanley Lab, Department of Bioinformatics and Computational Biology

Cross-platform immune signatures in ALS progression

- Showed that peripheral immune cell coordination patterns distinguish ALS patients from healthy donors across CyTOF and scRNA-seq cohorts more accurately than cell-type frequencies
- Developed computational approaches integrating multiple immune profiling platforms; code available on GitHub

Machine learning models for microglia aging prediction

- Developed ML models predicting biological age from single-cell microglia transcriptomic data, published in *Aging Cell*
- Created open-source software package (microglia-aging-clock) for broader scientific community use

Neuroimmune cell-cell communication in neurodegeneration

- Curating NeuroImmuneDB, a ligand-receptor database for neuroimmune cell-cell communication inference
- Analyzing cross-tissue communication across 6 scRNA-seq datasets (260K cells) to characterize neuroimmune axes altered in ALS
- Extending to compare signatures across neurodegenerative diseases (AD, MS, PD) and aging

Jun–Aug 2026 **Computational Biology in Neurodegeneration Intern** | Bristol Myers Squibb
Informatics and Predictive Sciences, Immunology, Cardiovascular & Neuroscience (ICVN), Early Neuroscience team

Disease progression modeling from fluid and imaging biomarkers in Alzheimer's disease

- Implemented and benchmarked published disease-progression modeling methods (SILA, TIRA) for estimating individual-level biomarker chronicity from longitudinal and cross-sectional Alzheimer's disease cohort data
- Compared model behavior and stability across plasma, CSF, and PET biomarker modalities

2021–2023 **Research Technician** | Weill Cornell Medical College
Hochrainer Lab, Brain & Mind Research Institute

Screening deubiquitinases for ischemic stress neuroprotection

- Transfected Neuro2A cells with siRNA for 96 deubiquitinases and imaged pre/post hydrogen peroxide treatment
- Developed ImageJ and R scripts for automated cell death quantification and candidate identification

PSD95-Tau interaction disruption to restore PSD95-nNOS coupling

- Conducted HEK293T transfections, western blotting, immunoprecipitation, and site-directed mutagenesis
- Investigated protein-protein interactions and disruption strategies

Kinase activity restoration following ischemic stroke ubiquitination

- Created CaMKII mutant plasmids and performed oxygen-glucose deprivation experiments on hippocampal slices
- Assessed ubiquitination, phosphorylation, and activity of PSD-localized kinases (CaMKII, CKβ, Pyk2) in MCAO mouse brain lysates, showing that removal of ubiquitin restores post-ischemic kinase activity
- Developed Shiny App for proteomic analysis, resulting in co-first author publication in *Communications Biology*

2018–2021 **Research Assistant** | Colgate University
Ingram Lab, Department of Biology

Circadian rhythms in Seasonal Affective Disorder

- Extracted and quantified RNA for PER3 and NR1D2 circadian genes from hair samples of 24 participants
- Assessed circadian differences by gender, mood disorder, and seasonal light exposure using MATLAB
- Published first-author paper in *Journal of Affective Disorders Reports*

COVID-19 pandemic impact on mental health and sleep

- Analyzed objective sleep data from Fitbit actigraphy and mental health surveys of 78 college students
- Compared pre-pandemic vs. pandemic data, contributing to publication in *PLOS ONE*

Latitudinal effects and polymorphisms in SAD and Depression

- Genotyped and screened 220 individuals from Upstate NY and Myanmar for circadian gene polymorphisms
- Conducted statistical analysis of associations between SAD, polymorphisms, gender, and population origin

Publications

Peer-Reviewed Publications

[5] **Stanley, N.***, **Dhawka, L.***, Jaikumar, S., Huang, Y.-C., & Zannas, A. S. (2025). Microglia Single-Cell RNA-Seq Enables Robust and Applicable Markers of Biological Aging. *Aging Cell*, e70095. https://doi.org/10.1111/acel.70095 (*co-first authorship)

[4] **Dhawka, L.***, **Palfini, V.***, Hambright, E., Blanco, I., Poon, C., Kahl, A., Resch, U., Bhawal, R., Benakis, C., Balachandran, V., Holder, A., Zhang, S., Iadecola, C., & Hochrainer, K. (2024). Post-ischemic ubiquitination at the postsynaptic density reversibly influences the activity of ischemia-relevant kinases. *Communications Biology*, *7*(1), 1–16. https://doi.org/10.1038/s42003-024-06009-8 (*co-first authorship)

[3] Dang, T., Russel, W. A., Saad, T., **Dhawka, L.**, Ay, A., & Ingram, K. K. (2023). Risk for Seasonal Affective Disorder (SAD) Linked to Circadian Clock Gene Variants. *Biology*, *12*(12), Article 12. https://doi.org/10.3390/biology12121532

[2] Knickerbocker, K. J., Cox, E. A., **Dhawka, L.**, Woods, K., & Ingram, K. K. (2022). Intra-individual impact of the COVID-19 pandemic on mental health and sleep in young adults. *PLoS ONE*, *17*(10), e0276165. https://doi.org/10.1371/journal.pone.0276165

[1] **Dhawka, L.**, Cha, Y., Ay, A., & Ingram, K. K. (2022). Low circadian amplitude and delayed phase are linked to seasonal affective disorder (SAD). *Journal of Affective Disorders Reports*, *10*, 100395. https://doi.org/10.1016/j.jadr.2022.100395

Preprints

[P2] Chen, C.-J., George, B., **Dhawka, L.**, Evangelista, B. A., & Stanley, N. (2026). MICRON learns outcome-associated representations of spatial immune microenvironments. *bioRxiv*, 2026.04.14.718488. https://doi.org/10.64898/2026.04.14.718488

[P1] **Dhawka, L.***, **Evangelista, B. A.***, Arooji, O. K., Iannone, M. A., Pellegrino, K., Traub, R., Li, X., Bedlack, R., Meeker, R. B., Cohen, T. J., & Stanley, N. (2025). Peripheral immune patterns enable robust cross-platform prediction of ALS onset and progression. *bioRxiv*, 2025.08.26.672381. https://doi.org/10.1101/2025.08.26.672381 (*co-first authorship)

In Acknowledgements

- Zafar, A., Overton, R., Attia, Z. Ay, A., & Ingram, K. K. (2022). Machine learning and expression analyses reveal circadian clock features predictive of anxiety. *Scientific Reports*, *12*, 5508. https://doi.org/10.1038/s41598-022-09421-4
- Weiss, C., Woods, K., Filipowicz, A., & Ingram, K. K. (2019). Sleep Quality, Sleep Structure, and PER3 Genotype Mediate Chronotype Effects on Depressive Symptoms in Young Adults. *Frontiers in Psychology.* https://doi.org/10.3389/fpsyg.2020.02028

Manuscripts in Preparation

- Holder, A., Hansen, A., **Dhawka, L.**, Palfini, V., Mithani, R., Hambright, E., Dannert, A., Mancinelli, C., Acosta, D. M., Racchumi, G., Li, T., Eliezer, D., Paquet, D., Iadecola, C., & Hochrainer, K. (2026). Tau-mediated postsynaptic dysfunction in tauopathy.
- Hambright, E., **Dhawka, L.**, Poon, C., Zhou, P., Blanco, I., Palfini, V., Kahl, A., Roth, A., Ghosh, A., Iadecola, C., & Hochrainer, K. (2026). Postsynaptic CaMKII activity after ischemic stroke is reversibly inhibited by CaMKIIa ubiquitination.

Software and Data

2026 **als-cross-platform-classification**: analysis code for cross-platform immune coordination models in ALS
github.com/ldhawka/als-cross-platform-classification
Companion annotated CyTOF and scRNA-seq datasets archived on Zenodo:
https://doi.org/10.5281/zenodo.16039495

2026	**Microglia Subtype Marker Atlas**: curated hierarchical reference of microglia and CNS-myeloid markers (9 umbrella families, 23 sub-states; paired human and mouse symbols, DOI-linked references) with a Python subtype-scoring tool; independent update of the marker table from `microglia-aging-clock` ldhawka.github.io/microglia-subtype-markers
2025	**microglia-aging-clock**: open-source package for generating single-cell microglia aging clocks github.com/CompCy-lab/microglia-aging-clock

Presentations

2026	Presenter, Computational Medicine Research in Progress Meeting, UNC Chapel Hill
2025	Presenter, Genetics/GMB/BCB Departmental Retreat, UNC Chapel Hill
2025	Presenter, Computational Medicine Research in Progress Meeting, UNC Chapel Hill
2024	Presenter, BCB New Student Symposium, UNC Chapel Hill
2023	Poster Presenter, UNC BBSP Fall Rotation Poster Session, UNC Chapel Hill
2021	Abstract and Virtual Poster, 85th Cold Spring Harbor Symposium – Biological Time Keeping
2020	Honors in Biology Thesis Defense Presentation, Colgate University
2019	Poster Presenter, Biology Senior Thesis Seminar Poster Session, Colgate University Poster Presenter, Summer Research Colloquium, Colgate University

Honors and Awards

2026	Kalman/AFAR Graduate Student Award in Geroscience, American Federation for Aging Research
2025	Best Talk Award, Genetics/GMB/BCB Departmental Retreat, UNC Chapel Hill
2020	Honors in Molecular Biology, Colgate University
2016–2020	Alumni Memorial Scholar awarded to top 10% of the incoming class, Colgate University
2017, 2018, 2019	Dean's Award with Distinction for academic excellence, Colgate University

Leadership, Teaching and Activities

2025–Present	**Co-director**, Human Functional Regulatory Genomics (HFROG) - trainee-focused discussion group, UNC Chapel Hill
2024–Present	**Editorial Coordinator, Writer, Editor**, The Pipette Pen, Science Writing and Communication Club, UNC Chapel Hill

2024	**Teaching Assistant**, BCB 720: Introduction to Statistical Modeling, UNC Chapel Hill
2017–2020	**Tutor**, Calculus II, Calculus III and Differential Equations, Center for Learning, Teaching, and Research, Colgate University
2017–2020	**Treasurer, President, Senior Advisor**, South Asian Cultural Club, Colgate University
2017–2018	**Secretary**, Hindu Students' Association, Colgate University
2016–2017	**Green Ambassador**, Office of Sustainability, Colgate University **Community Ambassador**, Hancock Residential Commons, Colgate University
2016–2018	**Volunteer**, Hamilton Food Cupboard

Technical Skills

Laboratory Techniques	DNA and RNA extraction and purification, rt-PCR, qPCR, microscopy, immunoprecipitation, gel electrophoresis, western blotting, cell culture, site-directed mutagenesis, mouse husbandry
Programming Languages	R, Python, MATLAB
Data Analysis	Single-cell RNA-seq analysis, CyTOF data analysis, machine learning, longitudinal and disease-progression modeling, bioinformatics pipeline development, ImageJ scripting, automated image analysis workflows
Software Development	GitHub version control, Snakemake workflow management, Shiny app development, open-source package creation
Software	Microsoft Office Suite, Google Drive, ImageJ, GraphPad Prism, SPSS Statistics

Work Experience

2018–2020 **Student Intern** | Office of Sustainability, Colgate University

- Conceived and piloted Oak Event Certification Program, promoting zero-waste campus initiatives
- Managed social media platforms and co-authored Colgate State of Sustainability Report 2019

2017–2018 **Residential Assistant** | Office of Residential Life, Colgate University

- Supervised first-year living area; presented with the 'Rise to the Occasion Award' for outstanding service
- Planned events with different departments to enrich the Liberal Arts College experience

Additional Qualifications

Languages	English (Fluent), French (Fluent), Hindi (Advanced), Mauritian Creole (native), Mandarin (Beginner)

Last updated: August 20, 2026